	Wikiplots					Writing Prompts					NY Times				
Model	AvgL	B-2	B-3	B-4	B-5	AvgL	B-2	B-3	B-4	B-5	AvgL	B-2	B-3	B-4	B-5
Gold Test	330	.74	.50	.29	.15	661	.82	.61	.40	.25	315	.73	.50	.32	.21
P&W-Static	352	.93	.85	.75	.64	675	.97	.94	.89	.85	352	.93	.85	.74	.63
Fusion	191	.84	.71	.58	.48	197	.93	.85	.75	.65	171	.89	.80	.70	.60
GROVER	835	.72	.49	.48	.37	997	.88	.72	.52	.34	719	.79	.57	.38	.25
GPT	909	.77	.47	.25	.11	799	**.73**	**.40**	**.19**	**.08**	739	**.68**	**.36**	.27	**.08**
GPT-2	910	.60	.26	.10	.03	799	.74	.41	**.19**	**.08**	756	.69	**.36**	**.17**	**.08**
PLOTMACHINES (GPT)	682	.77	.58	.40	.27	850	.89	.81	.72	.63	537	.85	.69	.53	.40
PLOTMACHINES (GPT-2)	553	**.56**	**.19**	**.07**	**.02**	799	.83	.56	.30	.14	455	.79	.57	.37	.23

Table 3: Average length of the generated test documents (**AvgL**) and Self-BLEU n-gram (**B-n**) scores on 1000 generated story samples from the test sets. We also include the average length and self-BLEU scores of the gold test data. A lower self-BLEU score together with a large ROUGE (see Table 2) score can justify the effectiveness of a model. We bold the lowest model score in each column; however, we note that sometimes the model self-bleu scores can be lower than in the gold document.

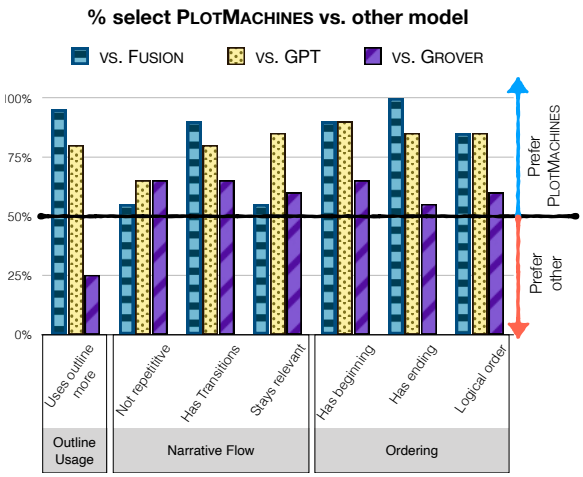

Figure 3: Head-to-head comparsion of PLOTMACHINES vs. three other models for full stories. In the appendix, we report results with standard error metrics (Table 9).

utilization, narrative flow, and ordering. We ask human raters[11] to evaluate each single or pair of generations from the Wikiplots test set, collecting 3 responses per story.[12]

To scale up the crowdsourcing work pragmatically, we split evaluations into two studies: one small-scale study evaluating full-length stories, and one large-scale study evaluating single-paragraph excerpts. In the first study, humans perform head-to-head ratings of 20 randomly sampled stories per pair of models. In the second study, humans rate story excerpts from 100 randomly sampled outputs per model.

[11]using Amazon Mechanical Turk. We include screenshots from the tasks Appendix A.2. In total, over 700 humans participated in all of our studies.

[12]For sampling from Fusion, we restrict to stories or excerpts with two or fewer unk tokens to ensure legibility for workers.

Outline Utilization		
Model A	Model B	% Prefer Model A
Random Paragraph		
PLOTMACHINES	Fusion	**80%** ±4.0
PLOTMACHINES	GPT	**72%** ±4.5
PLOTMACHINES	GROVER	49% ±5.0
Closest Paragraph		
PLOTMACHINES	Fusion	**83%** ±3.8
PLOTMACHINES	GPT	**83%** ±3.8
PLOTMACHINES	GROVER	**54%** ±5.0

Table 4: Humans judge which of two paragraphs better utilize the outlines (when shown either random paragraphs or the paragraphs most similar to the outline).

4.3.1 Full Story Ratings

We give human raters a pair of stories generated from the same outlines and ask them to choose which one is better in different aspects related to outline utilization, narrative flow, and ordering. In Figure 3, we show how often PLOTMACHINES (PM) was selected over the other models (values above 50% indicate that PM was preferred) using the majority vote for each example. PM was selected over base GPT and Fusion in all of the categories, demonstrating that the memory and discourse features are vitally important to improving the base model. While humans rated GROVER as using the outline more, PM is ranked higher in all of the questions about narrative flow and ordering.

4.3.2 Excerpt Ratings

Outline Usage We give raters two paragraphs each generated by different models and ask them to select which is utilizing the outline better. We perform two trials, one with random paragraphs from each story and one with the paragraph from each story that has the most n-gram overlap with the outline (i.e. the closest). In both cases, we compute the majority vote over the three responses and report the percentage of examples where our model is preferred. Results in Table 4 show that, when looking at single paragraphs, humans tend

Model	Narrative Flow			Order
	Rep(↓)	Tran(↑)	Rel(↑)	Acc(↑)
Fusion	2.61	2.98	3.36	**73**
GPT	**1.39**	1.89	2.06	42
GROVER	1.78	3.00	3.29	62
PM	1.64	**3.02**	**3.39**	59

Table 5: Human evaluations of paragraph excerpts from Fusion, GPT, GROVER and PLOTMACHINES (PM) outputs. Narrative flow questions rate the repetitiveness between paragraphs, transitioning, and relevance within paragraphs. In Table 10 of the appendix, we include standard error metrics.

to choose our PM as using the outlines in a more natural way, particularly when looking at the "closest" paragraph from both models. Fusion and GPT, in particular, are judged to be utilizing the outline much less than PLOTMACHINES.

Narrative Flow In this task, we give raters a generated paragraph (with the previous paragraph as context). They are asked to rate on a scale from 1 to 5 how much the paragraph: (a) repeats content from the previous paragraph, (b) transitions naturally from the previous paragraph, and (c) stays relevant and on-topic throughout the paragraph.

In the left side of Table 5, we show the average ratings of each model. GPT is the least repetitive between paragraphs but has very low subscores for transitions and relevance. We posit that this behavior is likely due to GPT often generating unrelated content from one paragraph to the next. PM tends to have the highest rated transitions and achieve highest relevancy within paragraphs while being much less repetitive between paragraphs than GROVER or Fusion.

Ordering It's challenging for humans to directly rate the ordering of a story based on a short excerpt. We instead set up a related proxy task: we give raters a pair of consecutive generated paragraphs, presented in a random order, and ask them to attempt to decipher the order. The intuition is that if the model output is very well-structured then it should be easier for humans to decipher the order. We compute the accuracy of the majority vote compared to the actual order in the right side of Table 5. Accuracy for PM approaches 60% accuracy and is much better than the base GPT. GROVER and Fusion are easiest for humans to re-order (62%, 73% respectively). This result differs slightly from the full story analysis where the humans preferred PM over GROVER and Fusion in the ordering-based questions. One possible explanation is that these two models, which decode word-by-word, without

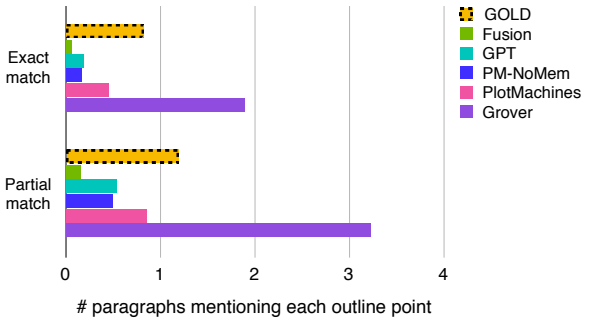

Figure 4: Number of paragraphs mentioning each outline point. PLOTMACHINES with memory covers points more similarly to the gold story, whereas GROVER tends to over-repeat outline points (twice as much as the gold reference).

an explicit notion of paragraph, may be better at resolving coreference problems between paragraphs. This may make it easier for humans to re-order short excerpts even though they generally prefer the overall narrative order of PM due to it having better beginnings, endings, etc. (as indicated in our full story human study).

4.4 N-gram Based Outline Usage Analysis

We perform an additional quantitative study to further investigate how outline points are used in generated stories. For fifty stories in the Wikiplots dev. set, we compute how many paragraphs mention each outline point using exact matching or partial matching ($> 20\%$ of the n-grams in the outline point also appear in the paragraph). We report the results in Figure 4.

We observe that GROVER tends to over-repeat outline points (about twice as much as the gold story). This mirrors our human evaluations that GROVER is more repetitive. This may also explain why human raters in the full story ratings in Sec. 4.3.1 judged GROVER as using the outline more but having worse narrative flow and order. Similar observations have been made about pretrained language models in See et al. (2019) that the models followed story prompts very closely but often copied too much compared to human writing.

In contrast, the Fusion model tends to leave out portions of the outline. This may reflect the way Fusion was originally designed – for use with a task using more abstract prompts as input. The GPT and PM-NOMEM models, while more inclusive than Fusion, are also likely to exclude outline points. The full PM model is generally more inclusive and more similar to the gold reference than the other models. The gold story mentions each outline point

in around one paragraph on average, indicating that there is an ideal balance between the more conservative coverage achieved by our model and the over-repetitive coverage of GROVER.

4.5 Qualitative Examples

In the Appendix, (Sec. A.3), we include examples of model outputs on the validation set with annotations for incorporated outline points. Examples indicate that GROVER often finishes the story and then starts a new story partway through the document. This may help explain why GROVER over-repeats outline points and why humans judge it to be more repetitive and less consistently relevant. In contrast, our model adheres more to a beginning-middle-ending structure.

We also look at examples of introduction and conclusion paragraphs generated by PLOTMACHINES, investigating the discourse the model has learned (Table 11). The model often starts stories by setting the scene (e.g. "In the early 1950s, a nuclear weapons testing continues ....") and tends to write conclusions with a definitive closing action (e.g. "... the film ends with humperdinck and buttercup riding off into the sunset.")

5 Related Work

State Tracking There is a plethora of work in state tracking for dialogue where memory states are updated after each utterance (Thomson and Young, 2010; Young et al., 2010; Lee, 2013; Chao and Lane, 2019). Similarly, SC-LSTMs (Wen et al., 2015) dynamically updated dialogue act representations as a form of sentence planning in spoken dialogue generation. Memory and entity networks (Henaff et al., 2017; Sukhbaatar et al., 2015) and neural checklists (Kiddon et al., 2016) also used similar methods for tracking entities for other tasks. We adapt these techniques for generating stories while tracking plot state that is updated after each paragraph. Our method of decoding paragraphs recurrently also draws on existing work in hierarchical decoding (Li et al., 2015; Shen et al., 2019), which similarly decodes in multiple levels of abstraction over paragraphs, sentences, and words.

Controllable Story Generation There has been a variety of work focusing on generating stories in plot-controllable, plan-driven, or constrained ways (e.g. (Riedl and Young, 2010; Fan et al., 2018; Peng et al., 2018; Jain et al., 2017; Lebowitz, 1987; Ippolito et al., 2019; Pérez y Pérez and Sharples, 2001)). Similar work in creative generation has conditioned on keywords for poetry generation (Yan, 2016; Ghazvininejad et al., 2016; Wang et al., 2016). Outline-conditioned generation is complementary to these tasks in that outlines provide more flexibility than very fine-grained srl-based, event-based, or graph-based plans (Fan et al., 2019; Martin et al., 2017; Harrison et al., 2017; Li et al., 2013) and more structured grounding than coarse-grained prompts (Fan et al., 2018; Xu et al., 2018) or ending goals (Tambwekar et al., 2019). Another similar task generates five line stories from five keywords (Peng et al., 2018; Yao et al., 2019). We generalize to a similar set-up for long-form narratives. Similar to many recent works in this area, we use seq2seq-based approaches, implemented using transformers. We further expand upon the modeling for the challenges specific to our task by using state tracking and applying discourse structure.

6 Conclusion

We present outline-conditioned story generation, a new task for generating stories from outlines representing key plot elements. We facilitate training by altering three datasets to include plot outlines as input for long story generation. In order to keep track of plot elements, we create PLOTMACHINES which generates paragraphs using a high-level discourse structure and a dynamic plot memory keeping track of both the outline and story. Quantitative analysis shows that PLOTMACHINES is effective in composing tighter narratives based on outlines compared to competitive baselines.

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